Modèles stochastiques en finance
Séance 6
Modèles à volatilité locale. Modèles hybrides de volatilité.

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Local volatility framework

Dupire's local volatility

Local volatility in a stochastic volatility framework

Related models and extensions

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Traders use the Black & Scholes formula... but not with a constant volatility

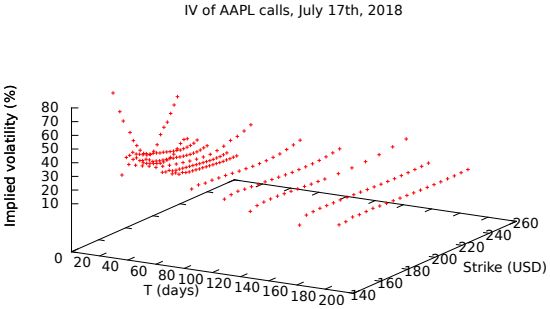

Figure: Implicit volatility (IV) of quoted call options on the underlying asset Apple (AAPL) as a function of strike and maturity, on July 17th, 2018. Data retrieved on `https://www.barchart.com/quotes/AAPL/options` on July 17th, 2018.

Local volatility framework

- Let $(\Omega, \mathcal{F}, \mathbf{P})$ be some probability space. Let $\overline{T}$ be a finite horizon. Let B be a $\mathbf{P}$-Brownian motions with filtration $(\mathcal{F}_t)_{t\in[0,\overline{T}]}$.
- In a general local local volatility framework, the dynamics of the asset price is written :

$$dS_t = \mu(t, S_t)S_t\, dt + \sigma(t, S_t)S_t\, dB_t.$$

- Any parametric form of the function $(t, x) \mapsto \sigma(t, x)$ defines a parametric local volatility model.
- Dupire (1994) proves that there exists a unique function $\sigma_L : [0, \overline{T}] \times \mathbb{R} \to \mathbb{R}_+^*$ such that the (vanilla) options prices generated by the model are consistent with the observed market prices.

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Kolmogorov forward equation

Theorem 6.1 (Kolmogorov forward, also Fokker-Planck)

Let $(\Omega, \mathcal{F}, \mathbf{P})$ be a probability space. Let B be a Brownian motion and $(\mathcal{F}_t)_{t\in[0,T]}$ its natural filtration. Let $(X_t)_{t\in[0,T]}$ be a stochastic process satisfying

$$dX_t = a(t, X_t)\, dt + b(t, X_t)\, dB_t,$$

where a and b satisfies the existence and uniqueness conditions. Let $s \in [0, T]$ and $x \in \mathbb{R}$. Let $p : [s, T] \times \mathbb{R} \to \mathbb{R}_+$, $(t, y) \mapsto p(t, y|s, x)$ be the conditional density of the process X with respect to the measure $\mathbf{P}$.
Then p is a solution to the partial differential equation

$$\frac{\partial}{\partial t} p(t, y|s, x) = -\frac{\partial}{\partial y}(a(t, y)p(t, y|s, x)) + \frac{1}{2}\frac{\partial^2}{\partial y^2}(b^2(t, y)p(t, y|s, x))$$

with initial condition $p(s, y|s, x) = \delta(y - x)$.

The Dupire approach I

- Let us assume that there exists a risk-free asset with constant rate $r > 0$ and introduce a martingale measure $\mathbf{Q}$ and include a continuous dividend rate $q > 0$ in the model. The dynamics is rewritten:

$$dS_t = (r - q)S_t\, dt + \sigma_L(t, S_t)S_t\, dB_t,$$

Theorem 6.2 (Dupire formula w.r.t. option prices)

Let $C(T, K)$ be the price at time t of the European call option with strike K and maturity $T < \overline{T}$. Then the function $\sigma_L : [0, \overline{T}] \times \mathbb{R} \to \mathbb{R}_+^*$ satisfies

$$\sigma_L^2(T, K) = \frac{\frac{\partial C}{\partial T}(T, K) + (r - q)K\frac{\partial C}{\partial K}(T, K) + qC(T, K)}{\frac{1}{2}K^2\frac{\partial^2 C}{\partial K^2}(T, K)}.$$

The Dupire approach II

- The previous result is known as the Dupire formula (Dupire, 1994).
- The model is theoretically consistent with observed prices, but calibration requires numerical derivatives of the observed surface of option prices (or implied variance, see next slide). Numerous related issues to be dealt with : liquidity, missing data, interpolation, smoothness, etc.

The Dupire formula w.r.t. implied volatility

- By inversion of the Black & Scholes formula, prices are quoted in terms of implied volatility, and the local volatility function can be expressed w.r.t. the implied variance surface.

Theorem 6.3 (Dupire formula w.r.t. the variance surface)

Let $q = 0$. Let $\xi = \log \frac{K}{S_0 e^{rT}}$ and $v_{BS}(T,K) = T\sigma_{BS}^2(T,K)$. The Dupire formula can be expressed w.r.t the implied variance surface :

$$\sigma_L^2(T,\xi) = \frac{\frac{\partial v_{BS}}{\partial T}}{1 - \frac{\xi}{v_{BS}}\frac{\partial v_{BS}}{\partial \xi} + \frac{1}{4}\left(-\frac{1}{4} - \frac{1}{v_{BS}} + \frac{\xi^2}{v_{BS}^2}\right)\left(\frac{\partial v_{BS}}{\partial \xi}\right)^2 + \frac{1}{2}\frac{\partial^2 v_{BS}}{\partial \xi^2}}$$

- More details in practitionners books, e.g. Gatheral (2006) or Bergomi (2016).

Dupire's forward equation

- Let us consider a given local volatility model:

$$dS_t = \mu(t, S_t)S_t\, dt + \sigma(t, S_t)S_t\, dB_t.$$

  for a given positive function $\sigma(t, x)$.
- Rewriting the Dupire formula we have the following partial differential equation:

$$\frac{\partial C}{\partial T}(T, K) + (r - q)K\frac{\partial C}{\partial K}(T, K) + qC(T, K) - \sigma^2(T, K)\frac{1}{2}K^2\frac{\partial^2 C}{\partial K^2}(T, K) = 0.$$

- Solving for this PDE for $(T, K) \in [0, \overline{T}] \times \mathbb{R}_+$ with initial condition $C(0, K) = (x - K)_+$ gives the call options prices for all maturities T and strikes K given $S_0 = x$ in this local volatility model.

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Interpreting local volatility w.r.t. stochastic volatility I

- Let us consider a general model of stochastic volatility under some risk-neutral measure $\mathbf{Q}$:

$$\begin{cases} dS_t = (r-q)S_t\,dt + \sqrt{v_t}S_t\,dZ_1(t), \\ dv_t = \alpha(t,S_t,v_t)\,dt + \beta(t,S_t,v_t)\,dZ_2(t). \end{cases}$$

  where Z_1 and Z_2 are correlated $\mathbf{Q}$-Brownian motions.

Proposition 6.4 (Local volatility in a stochastic volatility framework)

Computing Dupire's local volatility function with the option prices of the above stochastic volatility model yields the identity :

$$\sigma_L^2(t,x) = \mathbf{E}^{\mathbf{Q}}\left[v_t \middle| S_t = x\right].$$

Interpreting local volatility w.r.t. stochastic volatility II

- Local variance is the expectation w.r.t. the martingale measure $\mathbf{Q}$ of the *instantaneous* stochastic variance v_t, given S_t. See Derman and Kani (1998).
- Local volatility can thus be seen as a projection (conditional expectation) of a richer stochastic volatility framework.
- When dealing with complex/exotic/structured products, local volatility is often found to underprice products (skew too flat, missing vol-of-vol, ...)

♠ General result

- The above result can be linked to a general result on stochastic processes.

Theorem 6.5 (Gyöngy, 1986)

Let ξ_t be a stochastic process starting from 0 with Ito differential

$$d\xi_t = \alpha_t \, dt + \beta_t \, dW_t,$$

where W_t is a Brownian motion, and α and β are bounded adapted processes. There exists a stochastic differential equation

$$dX_t = a(t, X_t) \, dt + b(t, X_t) \, dW_t$$

where a and b are deterministic functions, with a weak solution X_t having the same one-dimensional probability distribution as ξ_t for every t. The coefficients a and b have a simple interpretation:

$$\begin{cases} a(t,x) = \mathbf{E}\left[\alpha_t \mid \xi_t = x\right], \\ b^2(t,x) = \mathbf{E}\left[\beta_t^2 \mid \xi_t = x\right]. \end{cases}$$

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CEV model

- The price of the risky asset solves the SDE
$$dS_t = rS_t\,dt + \alpha S_t^{1+\beta}\,dB_t,$$
with $\beta > -1$ constant.
- Can be interpreted as a local volatility model with $\sigma(t, S_t) = \alpha S_t^{\beta}$:
  - $\beta = 0$ is the Black & Scholes model ;
  - $\beta > 0$: ensure strictly positive asset prices, but volatility increases with prices ;
  - $-1 < \beta < 0$: volatility decreases with prices, but risk of bankruptcy.
- Well-known model because largely analytically tractable:
  - explicit transition densities (with Bessel functions) ; explicit probability of bankruptcy when $\beta < 0$;
  - explicit pricing formulas for vanilla options (with non-central chi-square distributions) ;
  - easy interpretation of the smile (α controls the level, β controls the slope) ;
  - analytical approximations of the implied volatility generated by a CEV model for the forward price.

Explicit call price in the CEV model

Theorem 6.6 (Explicit call price in the CEV model)

In a CEV model with dynamics $dS_t = rS_t\,dt + \delta S_t^{\beta/2}\,dB_t$, $\beta < 2$, *under an equivalent martingale measure, the price of a standard call option with strike* T *and horizon* T *is at time* t *with spot price* S_t*:*

$$C(t, St) = S_t Q(2y; 2 + 2/(2-\beta), 2x) - Ke^{-rt}(1 - Q(2x; 2/(2-\beta), 2y))$$

where $k = 2r/(\delta^2(2-\beta)(e^{r(2-\beta)(T-t)} - 1))$, $x = kS_t^{2-\beta}e^{r(2-\beta)(T-t)}$, $y = kK^{2-\beta}$ *and* $Q(x, a, b)$ *is the complementary cumulative distribution function of a non-central* ξ^2*-distribution with a degrees of freedom and non-centrality parameter* b*.*

- Mark Schroder, Computing the Constant Elasticity of Variance Option Pricing Formula, The Journal of Finance (1989) 44:1 pp.211–219.

The SABR model

- Used for example in interest rates modeling.
- Model definition : for $\beta \in [0,1]$, $\alpha > 0$, and two Brownian motions B^1 and B^2 such that $\langle B^1, B^2 \rangle_t = \rho t$, the model is written

$$\begin{cases} dX_t = \xi_t X_t^\beta dB_1(t), \\ d\xi_t = \alpha \xi_t dB_2(t), \end{cases}$$

  hence the name "Stochastic alpha beta rho" (SABR).
- Can be interpreted as local-stochastic volatility model with CEV local volatility.
- Popular model because of analytical approximations of the implied volatilities.
- To be used with care because of known inconsistencies due to the approximations.

Local-stochastic volatility models I

- Models mixing both a local and a stochastic component in the diffusion term:

$$\begin{cases} dS_t = rS_t\,dt + \sigma(t, S_t)\sqrt{\xi_t}S_t\,dB_t, \\ d\xi_t = \dots \end{cases}$$

- Using the previous interpretation, if $\sigma_L(T, K)$ is the local volatility function calibrated on the market smile, then (see above discussion) :

$$\sigma_L^2(t, x) = \mathbf{E}^{\mathbf{Q}}\left[\sigma^2(t, S_t)\xi_t \mid S_t = x\right],$$

hence

$$\sigma^2(t, x) = \frac{\sigma_L^2(t, x)}{\mathbf{E}^{\mathbf{Q}}\left[\xi_t \mid S_t = x\right]}.$$

- No simple solution since $\sigma(t, x)$ also plays a role in the right-hand side denominator.

Local-stochastic volatility models II

- Numerical solution via time discretization: Start with $\sigma(0,x) = \sigma_L(0,x)/\sqrt{\xi_0}$. If $\sigma(t_i,x)$ is known, we can use it to simulate N sample paths $(S^k, \xi^k)_{k=1,\dots,N}$ from t_i to t_{i+1} and compute the estimator :

$$\mathbf{E}^{\mathbf{Q}}\left[\xi_{t_{i+1}} \mid S_{t_{i+1}=x}\right] \approx \frac{\sum_{i=1}^N \xi_{t_{i+1}}^k h(S_{t_{i+1}}^k - x)}{\sum_{i=1}^N h(S_{t_{i+1}}^k - x)}$$

  with some regularizing kernel h (e.g., a Gaussian kernel). This gives $\sigma(t_{i+1},x)$ and then we iterate (particle method, Guyon and Henry-Labordère, 2012).
- Obviously, once the model is calibrated, Monte Carlo pricing of derivatives products is available.

References

- Bergomi, Lorenzo (2016). *Stochastic volatility modeling*. CRC Press.
- Derman, E. and I. Kani (1998). "Stochastic implied trees: Arbitrage pricing with stochastic term and strike structure of volatility". In: *International Journal of Theoretical and Applied Finance* 1, pp. 61–110.
- Dupire, Bruno (1994). "Pricing with a smile". In: *Risk* 7.1, pp. 18–20.
- Gatheral, Jim (2006). *The Volatility Surface: A Practitioner's Guide*. Wiley.
- Guyon, Julien and Pierre Henry-Labordère (2012). "Being particular about calibration". In: *Risk* 25.1, p. 88.
- Gyöngy, István (1986). "Mimicking the one-dimensional marginal distributions of processes having an Itô differential". In: *Probability theory and related fields* 71.4, pp. 501–516.
- Oksendal, Bernt (2013). *Stochastic differential equations: an introduction with applications*. Springer.